EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Minimum value of $\sin x \cos 2x$ is-

(A) 1 (B) –1 (C) $-\sqrt{\frac{2}{27}}$ (D) $-\frac{1}{3\sqrt{6}}$

MMM118

2. $f(x) = 1 + [\cos x]x$, in $0 \le x \le \frac{\pi}{2}$ (where [.] denotes greatest integer function)

(A) has a minimum value 0
(B) has a maximum value 2
(C) is continuous in $\left[0, \frac{\pi}{2}\right]$
(D) is not differentiable at $x = \frac{\pi}{2}$

MMM119

3. Minimum value of the function $f(x) = \sum_{k=1}^{5}(x-k)^2$ is at-

(A) $x = 2$ (B) $x = 5/2$ (C) $x = 3$ (D) $x = 5$

MMM120

4. Let $f(x) = \begin{bmatrix} \sin\frac{\pi x}{2} & , 0 \le x < 1 \\ 3 - 2x & , x \ge 1 \end{bmatrix}$ then :

(A) $f(x)$ has local maxima at $x = 1$
(B) $f(x)$ has local minima at $x = 1$
(C) $f(x)$ does not have any local extrema at $x = 1$
(D) $f(x)$ has a global minimum at $x = 1$

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MMM004

5. Values of a and b such that $f(x) = \frac{a}{x} + bx$ has a minimum at point (1, 6), are

(A) $a = 3, b = -3$ (B) $a = b = 3$ (C) $a = 6, b = 0$ (D) $a = -3, b = -3$

MMM005

6. If $f(x) = |x| + |x-1| + |x-2|$, then-

(A) f(x) has minima at $x = 1$
(B) $f(x)$ has maxima at $x = 0$
(C) Equation $f(x) = 1$ has 1 solution
(D) None of these

MMM010

7. If $f(x) = x^3 + ax^2 + bx + c$ is minimum at $x = 3$ and maximum at $x = -1$, then-

(A) $a = -3, b = -9, c = 0$
(B) $a = 3, b = 9, c = 0$
(C) $a = -3, b = -9, c \in R$
(D) none of these

MMM008

8. Range of the function $f(x) = \frac{\ell n x}{\sqrt{x}}$ is

(A) $(-\infty, e)$ (B) $(-\infty, e^2)$ (C) $\left(-\infty, \frac{2}{e}\right]$ (D) $\left(-\infty, \frac{1}{e}\right)$

MMM121

For More Join: @IITJEE_Advanced

9. If $(x-a)^{2m}(x-b)^{2n+1}$, where m and n are positive integers and $a > b$, is the derivative of a function f, then-
(A) $x = a$ gives neither a maximum, nor a minimum
(B) $x = a$ gives a maximum
(C) $x = b$ gives neither a maximum nor a minimum
(D) None of these

MMM122

10. If $'a'$ and $'b'$ are the greatest and least values of the function $f(x) = \cos x + \frac{1}{2}\cos 2x - \frac{1}{3}\cos 3x$, then

(A) $a-b = \frac{9}{4}$ (B) $a-b = \frac{9}{8}$ (C) $a-b = \frac{1}{12}$ (D) $a+b = \frac{9}{4}$

MMM001

11. The point(s) of minimum of the function, $f(x) = 4x^3 - x|x-2|, x \in [0,3]$ is:
(A) $x = 0$ (B) $x = 1/3$ (C) $x = 1/2$ (D) $x = 2$

MMM006

12. Difference between the greatest and the least values of the function $f(x) = x(\ell n\, x - 2)$ on $[1, e^2]$ is
(A) 2 (B) e (C) e^2 (D) 1

MMM009

13. $f(x)$ is cubic polynomial which has local maximum at $x = -1$, If $f(2) = 18$, $f(1) = -1$ and $f'(x)$ has local minima at $x = 0$, then
(A) the distance between $(-1, 2)$ and $(a, f(a))$, where $x = a$ is the point of local minima is $2\sqrt{5}$.
(B) $f(x)$ is increasing for $x \in [1, 2\sqrt{5})$
(C) $f(x)$ has local maxima at $x = 1$
(D) $f(0) = 5$

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MMM003

14. The cost of running a bus from A to B, is Rs. $\left(av + \frac{b}{v}\right)$, where v km/h is the average speed of the bus. When the bus travels at 30 km/h, the cost comes out to be Rs. 75 while at 40 km/h, it is Rs. 65. Then the most economical speed (in km/h) of the bus is :
(A) 40 (B) 60 (C) 45 (D) 50

MMM123

15. The sum of lengths of the hypotenuse and another side of a right angled triangle is given. The area of the triangle will be maximum if the angle between them is :
(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $5\pi/12$

MMM124

16. The bottom of the legs of a three legged table are the vertices of an isosceles triangle with sides 5, 5 and 6. The legs are to be braced at the bottom by three wires in the shape of a Y. The minimum length of the wire needed for this purpose, is
(A) $4 + 3\sqrt{3}$ (B) 10 (C) $3 + 4\sqrt{3}$ (D) $1 + 6\sqrt{2}$

MMM125

For More Join: @IITJEE_Advanced

17. Let $f(x)=\dfrac{\tan^n x}{1+\sum\limits_{r=1}^{2n}\tan^r x}$, $n\in N$, where $x\in\left[0,\dfrac{\pi}{2}\right)$

(A) $f(x)$ is bounded and it takes both of it's bounds and the range of $f(x)$ contains exactly one integral point.

(B) $f(x)$ is bounded and it takes both of it's bounds and the range of $f(x)$ contains more than one integral point.

(C) $f(x)$ is bounded but minimum and maximum does not exists.

(D) $f(x)$ is not bounded as the upper bound does not exist.

MMM126

18. The two parts of 100 for which the sum of double of first and square of second part is minimum, are

(A) 50, 50 (B) 99, 1 (C) 98, 2 (D) None of these

MMM002

19. The minimum value of the sum of sum and product of two numbers when their difference is '2' is

(A) 2 (B) – 2 (C) 0 (D) 1

MMM007

20. The set of all values of 'a' for which the function,

$f(x)=(a^2-3a+2)\left(\cos^2\dfrac{x}{4}-\sin^2\dfrac{x}{4}\right)+(a-1)x+\sin 1$ does not possess critical points is:

(A) $[1,\infty)$ (B) $(0,1)\cup(1,4)$ (C) $(-2,4)$ (D) $(1,3)\cup(3,5)$

MMM127

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For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Let $f(x)=\min\left\{|x|,\sqrt{1-x^2}\right\};x\in[-1,1]$ and $g(x)=\max\left\{|x|,\sqrt{1-x^2}\right\};x\in[-1,1]$.

Which of the following statement(s) is (are) correct ?

(A) number of local minima of $f(x)$ is 1

(B) number of local minima of $f(x)$ is 3

(C) $x=0$ is a point of local maxima for $g(x)$

(D) number of local maxima of $g(x)$ is 3

MMM033

2. If $f(x)=\dfrac{x}{1+x\tan x}, x\in\left(0,\dfrac{\pi}{2}\right)$, then

(A) $f(x)$ has exactly one point of minima

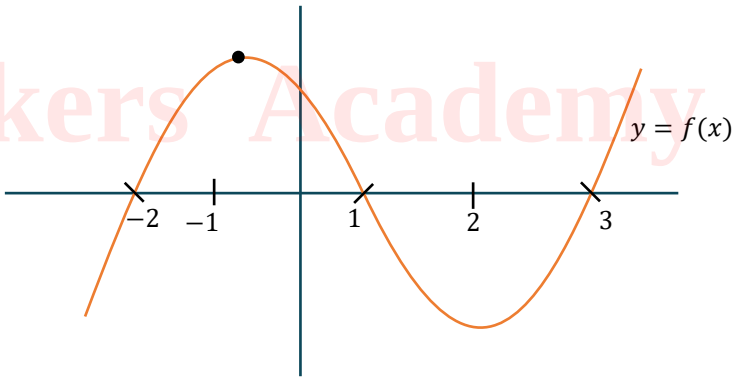
(B) $f(x)$ has exactly one point of maxima

(C) $f(x)$ is increasing in $\left(0,\dfrac{\pi}{2}\right)$

(D) maxima occurs at x_0 where $x_0 = \cos x_0$

MMM031

3. Graph of $y=f(x)$ is given, then



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(A) $y=H(x)=\max\{f(t):t\le x\}\ \forall\ x\in R$ has a local maximum as well as a local minimum

(B) $y=|f(x)|$ has 5 point of extrema

(C) $y=f(|x|)$ has 5 points of extrema

(D) $y=f(|x|)$ has 3 points of extrema

MMM029

4. Let $f(x,y)=\sqrt{x^2+y^2}+\sqrt{x^2+y^2-2x+1}+\sqrt{x^2+y^2-2y+1}+\sqrt{x^2+y^2-6x-8y+25}\ \forall x,y\in R$, then-

(A) Minimum value of $f(x,y)=5+\sqrt{2}$

(B) Minimum value of $f(x,y)=5-\sqrt{2}$

(C) Minimum value occurs of $f(x,y)$ for $x=\dfrac{3}{7}$

(D) Minimum value occurs of $f(x,y)$ for $y=\dfrac{4}{7}$

MMM030

For More Join: @IITJEE_Advanced

5. If the function $y = f(x)$ is represented as,

$x = \phi(t) = t^5 - 5t^3 - 20t + 7$

$y = \psi(t) = 4t^3 - 3t^2 - 18t + 3 \quad (|t| < 2)$, then :

(A) $y_{max} = 12$ (B) $y_{max} = 14$ (C) $y_{min} = -67/4$ (D) $y_{min} = -69/4$

MMM032

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 6 to 7

Let $f: R \to R, g: R \to R, h: R \to R$

$f(x) = x^2e^x, g(x) = x^3 - x^2 + x$ and $h(x) = x + 2$.

6. Choose the correct statement (s)

(A) The local minimum value of $f(g(x))$ is 0

(B) The local minimum value of $f(g(x))$ is an odd number

(C) The local minimum value of $f(g(x))$ does not exist

(D) The local minimum value of $f(g(x))$ is an even number

MMM037

7. Choose the correct statement (s)

(A) The local minimum value of $h(f(x))$ is 0

(B) The local minimum value of $h(f(x))$ is an odd number

(C) The local minimum value of $h(f(x))$ is 2

(D) The local minimum value of $h(f(x))$ is an even number

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MMM038

Paragraph for Question No. 8 to 10

Consider $f(x) = x^3(x-2)^2(x-1)$.

8. If least value of $f(x)$ is at $x = a$, then the value of $(3a - 2)$ is

MMM039

9. If least value of $f(x)$ is L, then the value of [L] is

(where [.] represents greatest integer function)

MMM040

10. If maxima and minima of $f(x)$ occurs at $x_i, i \in N$, then $\left[\sum_{i=1}^{n} x_i^2\right]$ is

(where [.] represents greatest integer function)

MMM041

Paragraph for Question No. 11 to 13

Let $f(x) = 1 + a^2x - x^3$ where 'a' is real number, point of minima of $f(x)$ must lie between $[A, B]$ where A and B is the minimum and maximum value of $\frac{x^2+3x+1}{x^2+x+1}$ for all $x \in R$.

On the basis of above information, answer the following questions-

For More Join: @IITJEE_Advanced

11. Find the value of A

(A) -1 (B) $-\frac{1}{2}$ (C) -2 (D) None of these

MMM034

12. Find the value of B

(A) $\frac{3}{2}$ (B) $\frac{5}{3}$ (C) $\frac{5}{4}$ (D) None of these

MMM035

13. If $a > 0$, then find the point of local minima

(A) $\frac{-a}{\sqrt{3}}$ (B) $\frac{a}{\sqrt{3}}$ (C) $\frac{a}{\sqrt{2}}$ (D) None of these

MMM036

MATRIX MATCH TYPE QUESTION

14.

	Column-I		**Column-II**
(A)	The number of point (s) of maxima of $f(x) = x^2 + \frac{1}{x^2}$ is	(P)	0
(B)	$(\sin^{-1}x)^3 + (\cos^{-1}x)^3$ is maximum at $x =$	(Q)	2
(C)	If $[a, b], (b < 1)$ is largest interval in which $f(x) = 3x^4 + 8x^3 - 6x^2 - 24x + 19$ is strictly increasing then $\frac{a}{b}$ is	(R)	$\frac{8}{3}$
(D)	If $a + b = 8, a, b > 0$ then minimum value of $\frac{a^3 + b^3}{48}$ is	(S)	-1

MMM042

15.

	Column-I		**Column-II**
(A)	A rectangle is inscribed in an equilateral triangle of side $4cm$. Square of maximum area of such a rectangle is	(P)	65
(B)	The volume of a rectangular closed box is 72 and the base sides are in the ratio 1 : 2. The least total surface area is	(Q)	45
(C)	If x and y are two positive numbers such that $x + y = 60$ and x^3y is maximum then value of x is	(R)	12
(D)	The sides of a rectangle of greatest perimeter which is inscribed in a semicircle of radius $\sqrt{5}$ are a and b. Then $a^3 + b^3 =$	(S)	108

MMM043

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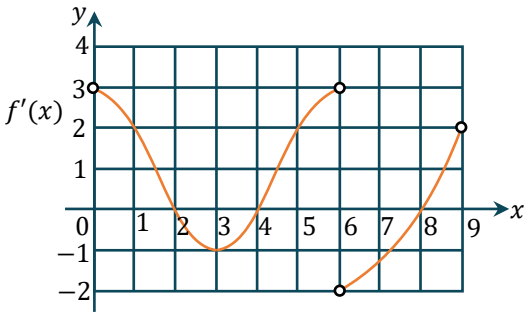
For More Join: @IITJEE_Advanced

EXERCISE (S)

1. The maximum value of function $f(x) = x^3 - 12x^2 + 36x + 17$ in the interval $[1, 10]$ is

MMM011

2. The graph of the derivative $f'(x)$ of a continuous function $f(x)$ in $(0, 9)$. If



(i) f is strictly increasing in the interval
$(a, b]$; $[c, d]$; $[e, f)$ and strictly decreasing in $[p, q]$; $[r, s]$.
(ii) f has a local minima at $x = x_1$ and $x = x_2$.
(iii) $f''(x) > 0$ in (l, m) ; (n, t)
(iv) f has inflection point at $x = k$
(v) number of critical points of $y = f(x)$ is 'w'.
Find the value of $(a + b + c + d + e) + (p + q + r + s) + (l + m + n) + (x_1 + x_2) + (k + w)$.

MMM044

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3. Let $P(x)$ be a polynomial of degree 5 having extremum at $x = -1, 1$ and $\lim\limits_{x\to 0}\left(\frac{P(x)}{x^3} - 2\right) = 4$. If M and m are the maximum and minimum value of the function $y = P'(x)$ on the set $A = \{x | x^2 + 6 \le 5x\}$ then find $\frac{m}{M}$.

MMM045

4. Maximum possible area that can be enclosed by a wire of length 20 cm by bending it in form of a circular sector is

MMM012

5. Two vertices of a rectangle are on the positive x-axis. The other two vertices lie on the lines $y = 4x$ and $y = -5x + 6$. Then the maximum area of the rectangle is A then find $100A$

MMM013

6. P is a point on positive x-axis, Q is a point on the positive y-axis and 'O' is the origin. If the line passing through P and Q is tangent to the curve $y = 3 - x^2$ then the minimum area of the triangle OPQ, is

MMM014

7. There are 50 apple trees in an orchard. Each tree produces 800 apples. For each additional tree planted in the orchard, per tree the output drop by 10 times the number of additional trees. Number of trees that should be added to the existing orchard for maximising the output of the trees, is

MMM015

For More Join: @IITJEE_Advanced

8. Of all the lines tangent to the graph of the curve $y = \frac{6}{x^2+3}$, then the sum of the minimum and maximum slopes of these tangent lines.

MMM046

9. A running track of 440 $ft.$ is to be laid out enclosing a football field, the shape of which is a rectangle with semi circle at each end. If the area of the rectangular portion is to be maximum, then the perimeter of the rectangular portion is
Use : $\pi \approx 22/7$.

MMM047

10. The value of 'a' for which $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ have a positive point of maximum lies in the interval $(a_1, a_2) \cup (a_3, a_4)$. Find the value of $a_2 + 11a_3 + 70a_4$.

MMM048

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. A helicopter is flying along the curve given by $y - x^{\frac{3}{2}} = 7$, $(x \geq 0)$. A soldier positioned at the point $\left(\frac{1}{2}, 7\right)$ wants to shoot down the helicopter when it is nearest to him. Then this nearest distance is :

[JEE (Main) 2019]

(1) $\frac{1}{2}$ (2) $\frac{1}{3}\sqrt{\frac{7}{3}}$ (3) $\frac{1}{6}\sqrt{\frac{7}{3}}$ (4) $\frac{\sqrt{5}}{6}$

MMM018

2. Let $a_1, a_2, a_3, \ldots\ldots$be an A. P. with $a_6 = 2$. Then the common difference of this A. P., which maximises the produce $a_1 a_4 a_5$, is : **[JEE (Main) 2019]**

(1) $\frac{6}{5}$ (2) $\frac{8}{5}$ (3) $\frac{2}{3}$ (4) $\frac{3}{2}$

MMM019

3. Let $f(x)$ be a polynomial of degree 5 such that $x = \pm 1$ are its critical points. If $\lim\limits_{x \to 0}\left(2 + \frac{f(x)}{x^3}\right) = 4$, then which one of the following is not true? **[JEE (Main) 2020]**

(1) f is an odd function

(2) $x = 1$ is a point of minima and $x = -1$ is a point of maxima of f.

(3) $x = 1$ is a point of maxima and $x = -1$ is a point of minimum of f.

(4) $f(1) - 4f(-1) = 4$

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MMM020

4. Let $f(x)$ be a polynomial of degree 3 such that $f(-1) = 10, f(1) = -6, f(x)$ has a critical point at $x = -1$ and $f'(x)$ has a critical point at $x = 1$. Then $f(x)$ has a local minima at $x =$ _____.

[JEE (Main) 2020]

MMM021

5. A spherical iron ball of 10 cm radius is coated with a layer of ice of uniform thickness the melts at a rate of 50 cm^3/min. When the thickness of ice is 5 cm, then the rate (in cm/min.) at which of the thickness of ice decreases, is : **[JEE (Main) 2020]**

(1) $\frac{1}{36\pi}$ (2) $\frac{5}{6\pi}$ (3) $\frac{1}{18\pi}$ (4) $\frac{1}{54\pi}$

MMM022

6. The triangle of maximum area that can be inscribed in a given circle of radius 'r' is :

(1) An isosceles triangle with base equal to $2r$. **[JEE (Main) 2021]**

(2) An equilateral triangle of height $\frac{2r}{3}$.

(3) An equilateral triangle having each of its side of length $\sqrt{3}r$.

(4) A right angle triangle having two of its sides of length $2r$ and r.

MMM023

For More Join: @IITJEE_Advanced

7. The sum of all the local minimum values of the twice differentiable function $f:R \to R$ defined by $f(x)=x^3-3x^2-\frac{3f''(2)}{2}x+f''(1)$ is : **[JEE (Main) 2021]**

(1) –22 (2) 5 (3) –27 (4) 0

MMM024

8. The sum of the absolute minimum and the absolute maximum values of the function $f(x) = |3x - x^2 + 2| - x$ in the interval $[-1, 2]$ is : **[JEE (Main) 2021]**

(1) $\frac{\sqrt{17}+3}{2}$ (2) $\frac{\sqrt{17}+5}{2}$ (3) 5 (4) $\frac{9-\sqrt{17}}{2}$

MMM025

9. The curve $y(x) = ax^3 + bx^2 + cx + 5$ touches the x-axis at the point $P(-2, 0)$ and cuts the y-axis at the point Q, where y' is equal to 3. Then the local maximum value of $y(x)$ is : **[JEE (Main) 2022]**

(1) $\frac{27}{4}$ (2) $\frac{29}{4}$ (3) $\frac{37}{4}$ (4) $\frac{9}{2}$

MMM026

10. A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1}\frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____. **[JEE (Main) 2022]**

MMM027

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11. Let $x = 2$ be a local minima of the function $f(x)=2x^4-18x^2+8x+12$, $x\in(-4,4)$. If M is local maximum value of the function f in $(-4, 4)$, then M = **[JEE (Main) 2023]**

(1) $12\sqrt{6}-\frac{33}{2}$ (2) $12\sqrt{6}-\frac{31}{2}$ (3) $18\sqrt{6}-\frac{33}{2}$ (4) $18\sqrt{6}-\frac{31}{2}$

MMM028

12. A square piece of tin of side 30 cm is to be made into a box without top by cutting a square from each corner and folding up the flaps to form a box. If the volume of the box is maximum, then its surface area (in cm^2) is equal to **[JEE (Main) 2023]**

(1) 675 (2) 1025 (3) 800 (4) 900

MMM067

13. $\max\limits_{0\le x\le\pi}\left\{x-2\sin x\cos x+\frac{1}{3}\sin 3x\right\}=$ **[JEE (Main) 2023]**

(1) $\frac{5\pi+2+3\sqrt{3}}{6}$ (2) $\frac{\pi+2-3\sqrt{3}}{6}$ (3) π (4) 0

MMM068

14. If the functions $f(x)=\frac{x^3}{3}+2bx+\frac{ax^2}{2}$ and $g(x)=\frac{x^3}{3}+ax+bx^2, a\neq 2b$ have a common extreme point, then $a + 2b + 7$ is equal to **[JEE (Main) 2023]**

(1) 4 (2) $\frac{3}{2}$ (3) 3 (4) 6

MMM069

For More Join: @IITJEE_Advanced

15. Let $f(x)=3\sqrt{x-2}+\sqrt{4-x}$ be a real valued function. If α and β are respectively the minimum and the maximum values of f, then $\alpha^2 + 2\beta^2$ is equal to **[JEE (Main) 2024]**

(1) 44 (2) 42 (3) 24 (4) 38

MMM128

16. Let $f(x) = 4\cos^3 x + 3\sqrt{3}\cos^2 x - 10$. The number of points of local maxima of f in interval $(0, 2\pi)$ is: **[JEE (Main) 2024]**

(1) 1 (2) 2 (3) 3 (4) 4

MMM129

17. Let the set of all values of p, for which $f(x) = (p^2 - 6p + 8)(\sin^2 2x - \cos^2 2x) + 2(2 - p)x + 7$ does not have any critical point, be the interval (a, b). Then $16ab$ is equal to ____ . **[JEE (Main) 2024]**

MMM130

18. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1, a > 0$ has a local maximum at $x=\alpha$ and a local minimum $x=\alpha^2$, then α and α^2 are the roots of the equation : **[JEE (Main) 2024]**

(1) $x^2-6x+8=0$ (2) $8x^2+6x-8=0$ (3) $8x^2-6x+1=0$ (4) $x^2+6x+8=0$

MMM131

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Let f, g and h be real-valued functions defined on the interval [0, 1] by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ and $h(x) = x^2e^{x^2} + e^{-x^2}$. If a, b and c denote respectively, the absolute maximum of f, g and h on [0, 1], then

(A) $a = b$ and $c \neq b$ (B) $a = c$ and $a \neq b$ (C) $a \neq b$ and $c \neq b$ (D) $a = b = c$

[JEE (Advanced) 2010]

MMM049

2. Let f be a function defined on $\boldsymbol{R}$ (the set of all real numbers) such that
$f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$, for all $x \in R$.
If g is a function defined on R with values in the interval $(0,\infty)$ such that
$f(x) = \ell n(g(x))$, for all $x \in R$,
then the number of points in R at which g has a local maximum is

[JEE (Advanced) 2010]

MMM050

3. Let $f: IR \to IR$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is **[JEE (Advanced) 2012]**

MMM051

4. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x = 1$ and a local minimum at $x = 3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is **[JEE (Advanced) 2012]**

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MMM052

5. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio of 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of the sides of the rectangular sheet are **[JEE (Advanced) 2013]**

(A) 24 (B) 32 (C) 45 (D) 60

MMM053

6. The function $f(x) = 2|x| + |x + 2| - ||x + 2| - 2|x||$ has a local minimum or a local maximum at $x =$ **[JEE (Advanced) 2013]**

(A) –2 (B) $\frac{-2}{3}$ (C) 2 (D) $\frac{2}{3}$

MMM054

7. A cylindrical container is to be made from certain solid material with the following constraints. It has a fixed inner volume of $V\ mm^3$, has a 2 mm thick solid wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container.

If the volume of the material used to make the container is minimum when the inner radius of the container is $10mm$, then the value of $\frac{V}{250\pi}$ is **[JEE (Advanced) 2015]**

MMM055

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8. The least value of $\alpha \in \mathbb{R}$ for which $4\alpha x^2 + \frac{1}{x} \geq 1$, for all $x > 0$, is - **[JEE (Advanced) 2016]**

(A) $\frac{1}{64}$ (B) $\frac{1}{32}$ (C) $\frac{1}{27}$ (D) $\frac{1}{25}$

MMM056

9. Let $f: \mathbb{R} \to (0, \infty)$ and $g: \mathbb{R} \to \mathbb{R}$ be twice differentiable function such that f'' and g'' are continuous functions on $\mathbb{R}$. Suppose $f'(2) = g(2) = 0$, $f''(2) \neq 0$ and $g'(2) \neq 0$. If $\lim\limits_{x \to 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$, then

[JEE (Advanced) 2016]

(A) f has a local minimum at $x = 2$
(B) f has a local maximum at $x = 2$
(C) $f''(2) > f(2)$
(D) $f(x) - f''(x) = 0$ for at least one $x \in \mathbb{R}$

MMM057

10. If $f(x) = \begin{vmatrix} \cos(2x) & \cos(2x) & \sin(2x) \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$, then **[JEE (Advanced) 2017]**

(A) $f'(x) = 0$ at exactly three points in $(-\pi, \pi)$
(B) $f(x)$ attains its maximum at $x = 0$
(C) $f(x)$ attains its minimum at $x = 0$
(D) $f'(x) = 0$ at more than three points in $(-\pi, \pi)$

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MMM058

11. For every twice differentiable function $f: \mathbb{R} \to [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, which of the following statement(s) is (are) TRUE ? **[JEE (Advanced) 2018]**

(A) There exist $r, s \in \mathbb{R}$, where $r < s$, such that f is one-one on the open interval (r, s).
(B) There exists $x_0 \in (-4, 0)$ such that $|f'(x_0)| \leq 1$
(C) $\lim\limits_{x \to \infty} f(x) = 1$
(D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \neq 0$

MMM059

12. Let $f(x) = \frac{\sin \pi x}{x^2}$, $x > 0$

Let $x_1 < x_2 < x_3 < \ldots < x_n < \ldots$ be all the points of local maximum of f
and $y_1 < y_2 < y_3 < \ldots < y_n < \ldots$ be all the points of local minimum of f.
Then which of the following options is/are correct ? **[JEE (Advanced) 2019]**

(1) $|x_n - y_n| > 1$ for every n (2) $x_1 < y_1$

(3) $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ for every n (4) $x_{n+1} - x_n > 2$ for every n

MMM060

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13. Consider all rectangles lying in the region

$$\left\{(x,y)\in\mathbb{R}\times\mathbb{R}:0\le x\le\frac{\pi}{2}\text{ and }0\le y\le 2\sin(2x)\right\}$$

and having one side on the x-axis. The area of the rectangle which has the maximum perimeter among all such rectangles, is **[JEE (Advanced) 2020]**

(A) $\frac{3\pi}{2}$ (B) π (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi\sqrt{3}}{2}$

MMM061

14. Let the function $f:(0,\pi)\to\mathbb{R}$ be defined by

$f(\theta)=(\sin\theta+\cos\theta)^2+(\sin\theta-\cos\theta)^4$

Suppose the function f has a local minimum at θ precisely when $\theta\in\{\lambda_1\pi,\ldots,\lambda_r\pi\}$, where $0<\lambda_1<\ldots.<\lambda_r<1$. Then the value of $\lambda_1+\ldots+\lambda_r$ is ____ **[JEE (Advanced) 2020]**

MMM062

Comprehension for Questions Nos. 15 and 16

Let $f_1:(0,\infty)\to\mathbb{R}$ and $f_2:(0,\infty)\to\mathbb{R}$ be defined by $f_1(x)=\int_0^x\prod_{j=1}^{21}(t-j)^j\,dt$, $x>0$

and

$f_2(x)=98(x-1)^{50}-600(x-1)^{49}+2450, x>0,$

where, for any positive integer n and real numbers $a_1,a_2,\ldots,a_n$, $\prod_{i=1}^{n}a_i$ denotes the product of $a_1,a_2,\ldots,a_n$. Let m_i and n_i, respectively, denote the number of points of local minima and the number of points of local maxima of function $f_i, i=1,2$, in the interval $(0,\infty)$

15. The value of $2m_1+3n_1+m_1n_1$ is ____. **[JEE (Advanced) 2021]**

MMM063

16. The value of $6m_2+4n_2+8m_2n_2$ is ____. **[JEE (Advanced) 2021]**

MMM064

17. Let $\alpha=\sum_{k=1}^{\infty}\sin^{2k}\left(\frac{\pi}{6}\right)$.

Let $g:[0,1]\to\mathbb{R}$ be the function defined by

$g(x)=2^{\alpha x}+2^{\alpha(1-x)}$

Then, which of the following statements is/are TRUE ? **[JEE (Advanced) 2022]**

(A) The minimum value of $g(x)$ is $2^{\frac{7}{6}}$

(B) The maximum value of $g(x)$ is $1+2^{\frac{1}{3}}$

(C) The function $g(x)$ attains its maximum at more than one point

(D) The function $g(x)$ attains its minimum at more than one point

MMM065

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18. Let $f:(0,1)\to\mathbb{R}$ be the function defined as $f(x)=[4x]\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right)$, where $[x]$ denotes the greatest integer less than or equal to x. Then which of the following statements is(are) true?

(A) The function f is discontinuous exactly at one point in $(0,1)$

(B) There is exactly one point in $(0,1)$ at which the function f is continuous but **NOT** differentiable

(C) The function f is **NOT** differentiable at more than three points in $(0,1)$

(D) The minimum value of the function f is $-\frac{1}{512}$

[JEE (Advanced) 2023]

MMM132

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	A	B	A	C	C	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	B	B	C	A	A	B	B	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C,D	B,D	A,B,D	A,C,D	B,D	A,D	C,D	0	–1	6
Que.	11	12	13	14			15			
Ans.	A	B	A	A→P; B→S; C→Q; D→R			A→R; B→S; C→Q; D→P			

EXERCISE (S)

1.	177	**2.**	74	**3.**	6	**4.**	25	**5.**	80
6.	4	**7.**	15	**8.**	0	**9.**	360	**10.**	320

EXERCISE - JEE (Main) PYQ

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Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	2	3	3	3	3	1	1	5
Que.	11	12	13	14	15	16	17	18		
Ans.	1	3	1	4	2	2	252	1		

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	1	5	9	A,C	A,B	4	C	A,D	B,D
Que.	11	12	13	14	15	16	17	18		
Ans.	A,B,D	A,C,D	C	0.50	57	6	A,B,C	A,B		

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Important Notes

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